Taehyun (Brian) Lee

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EDUCATION

University of Western Ontario Sept. 2024 – Present
Bachelor of Science (BSc) – Computer Science London, ON
GPA: 3.9/4.0

EXPERIENCE

Western Cyber Society Oct. 2025 – Apr. 2026

Skin Cancer Diagnostics (Python, PyTorch, Next.js, FastAPI) London, ON

- Engineered an AI-driven skin cancer diagnostics application by optimizing a CNN model to classify 10,000+ images with **91% accuracy**.
- Utilized data augmentation techniques such as MixUp to reduce overfitting and AMP to **speed up training by 34%** with lower GPU memory use.
- Improved user experience by integrating **real-time LLM** and exporting data trends with confidence score as pdf.

PROJECT

DeskCanvas *(C#)*

- **Published a Windows desktop customization app on Steam** with configurable taskbars, desktop widgets, themes, and animated popups.
- Engineered **multi-monitor overlay and window-management systems** with display-aware placement, full screen handling, and integrations for system status, notifications, audio, and media controls.
- Built visual editors and preset workflows for customizing and saving layouts with **Steam Workshop integration** for sharing and browsing presets within community.

Tower Defence Game *(Python, PyGame)*

- Created a full-featured tower defence game with object-oriented systems for enemies, towers, upgrades, and UI menus.
- **Designed and optimized gameplay algorithms** for enemy pathfinding, collision detection, attack logic, and upgrade scaling, balancing progression across 15 waves.
- **Enhanced user interaction** by incorporating game controls, difficulty settings, and real-time audio-visual feedback.

Tetris *(Python, PyGame)*

- Built a Python-based recreation of Tetris with core systems for **block spawning, rotation algorithms, collision detection, and advanced scoring logic**.
- Implemented custom gameplay features including **score multipliers, bonus rewards, and progressive difficulty**, extending beyond standard line-clearing mechanics.
- Integrated multiple play modes including infinite mode, 40-line challenge, and two-player competitive gameplay.

Scientific Calculator *(Python)*

- Developed a GUI-based scientific calculator in Python using Tkinter, supporting arithmetic, trigonometric, logarithmic, factorial, and square root operations.
- **Applied algorithmic logic for expression parsing**, including list-based operator storage, bracket balancing, and evaluation flow control to handle complex inputs reliably.
- Architected advanced features such as unit conversions, radians/degrees switching, and shift functions, extending functionality beyond standard calculators.

TECHNICAL SKILL

Languages: Python, Java, C#, C, JavaScript, TypeScript, HTML/CSS, R
Libraries/Frameworks: PyTorch, NumPy, React, Next.js, FastAPI, Tailwind CSS, PyGame
Tools: Git, GitHub, VS Code, Jupyter Notebook, Unix, Vite